

Git Branching

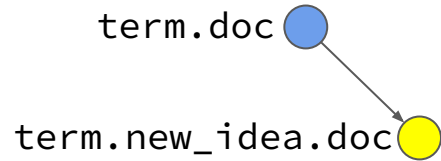
Software Engineering for Scientists

A **branch** is an independent line of development
(new file, changes to files)

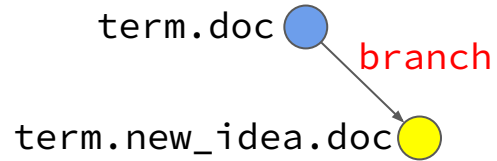
A **branch** is an independent line of development
(new file, changes to files)

term.doc 

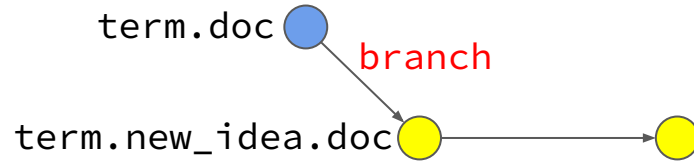
A **branch** is an independent line of development
(new file, changes to files)



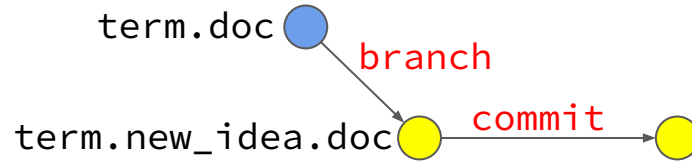
A **branch** is an independent line of development
(new file, changes to files)



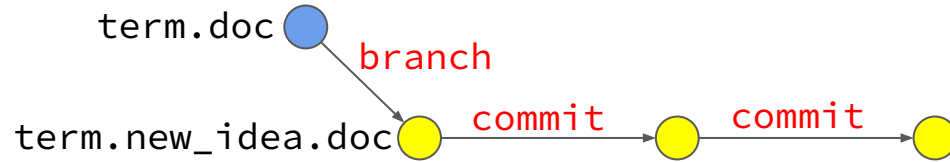
A **branch** is an independent line of development
(new file, changes to files)



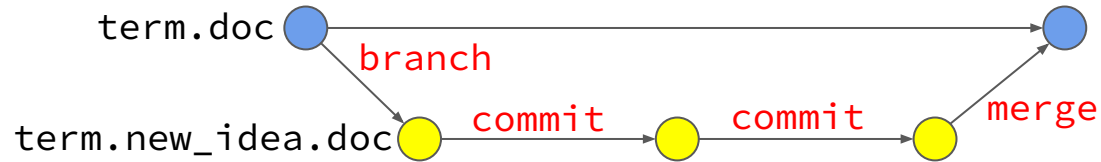
A **branch** is an independent line of development
(new file, changes to files)



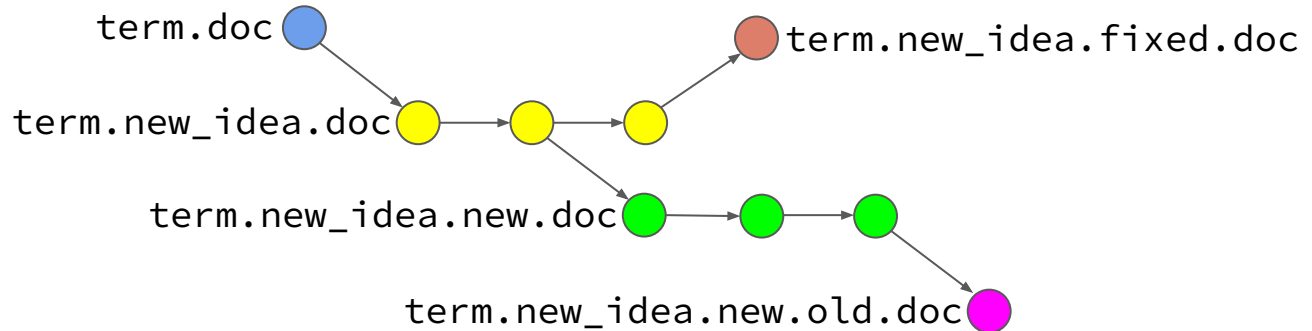
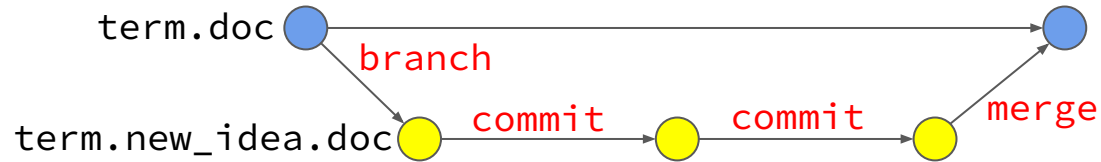
A **branch** is an independent line of development
(new file, changes to files)



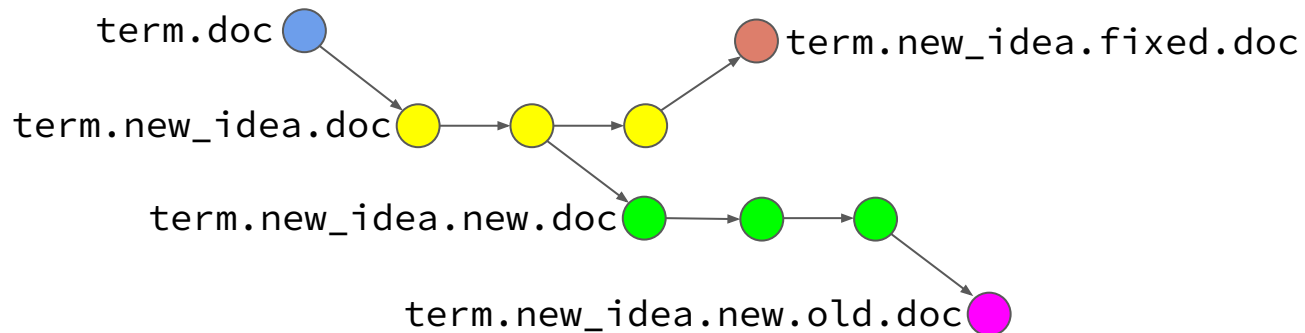
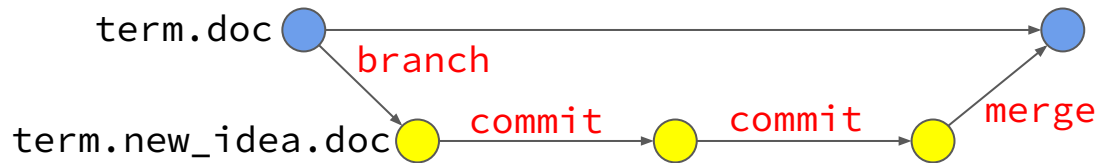
A **branch** is an independent line of development
(new file, changes to files)



A **branch** is an independent line of development
(new file, changes to files)



A **branch** is an independent line of development
(new file, changes to files)



git branching/merging is cheap and fast

github

remote repository

lib.py

```
def div(a, b):  
    return a/b
```

clone

Alice

local repository

local workspace

lib.py

```
def div(a, b):  
    if b > 0:  
        return a/b  
    else:  
        return None
```

push

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b > 0:  
        return a/b  
    else:  
        return None
```

lib.py

```
def div(a, b):  
    if b > 0:  
        return a/b  
    else:  
        return None
```

commit

github

remote repository

lib.py

```
def div(a, b):  
    return a/b
```

github

remote repository

branch → main

lib.py

```
def div(a, b):  
    return a/b
```

github

remote repository

main

lib.py

```
def div(a, b):  
    return a/b
```

clone

local repository

Alice

main

lib.py

```
def div(a, b):  
    return a/b
```

github

remote repository

main

lib.py

```
def div(a, b):  
    return a/b
```

clone

Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

branch

fix_div

lib.py

```
def div(a, b):  
    return a/b
```


github

remote repository

main

lib.py

```
def div(a, b):  
    return a/b
```

clone

Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

branch

fix_div

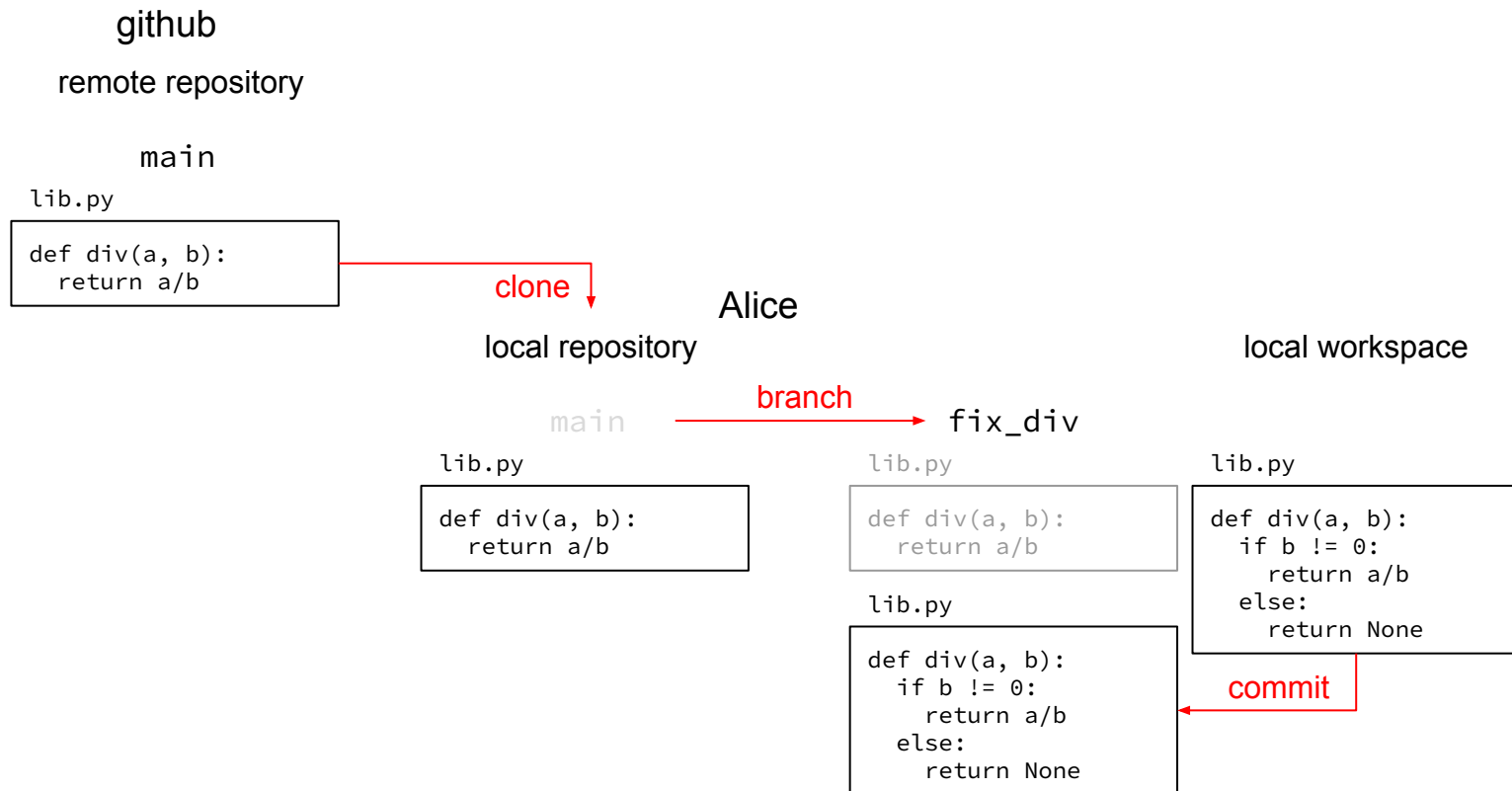
lib.py

```
def div(a, b):  
    return a/b
```

local workspace

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```



github

remote repository

main

lib.py

```
def div(a, b):  
    return a/b
```

clone

Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

branch

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

local workspace

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

commit

merge

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

github

remote repository

master

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

clone

Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

branch

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

local workspace

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

commit

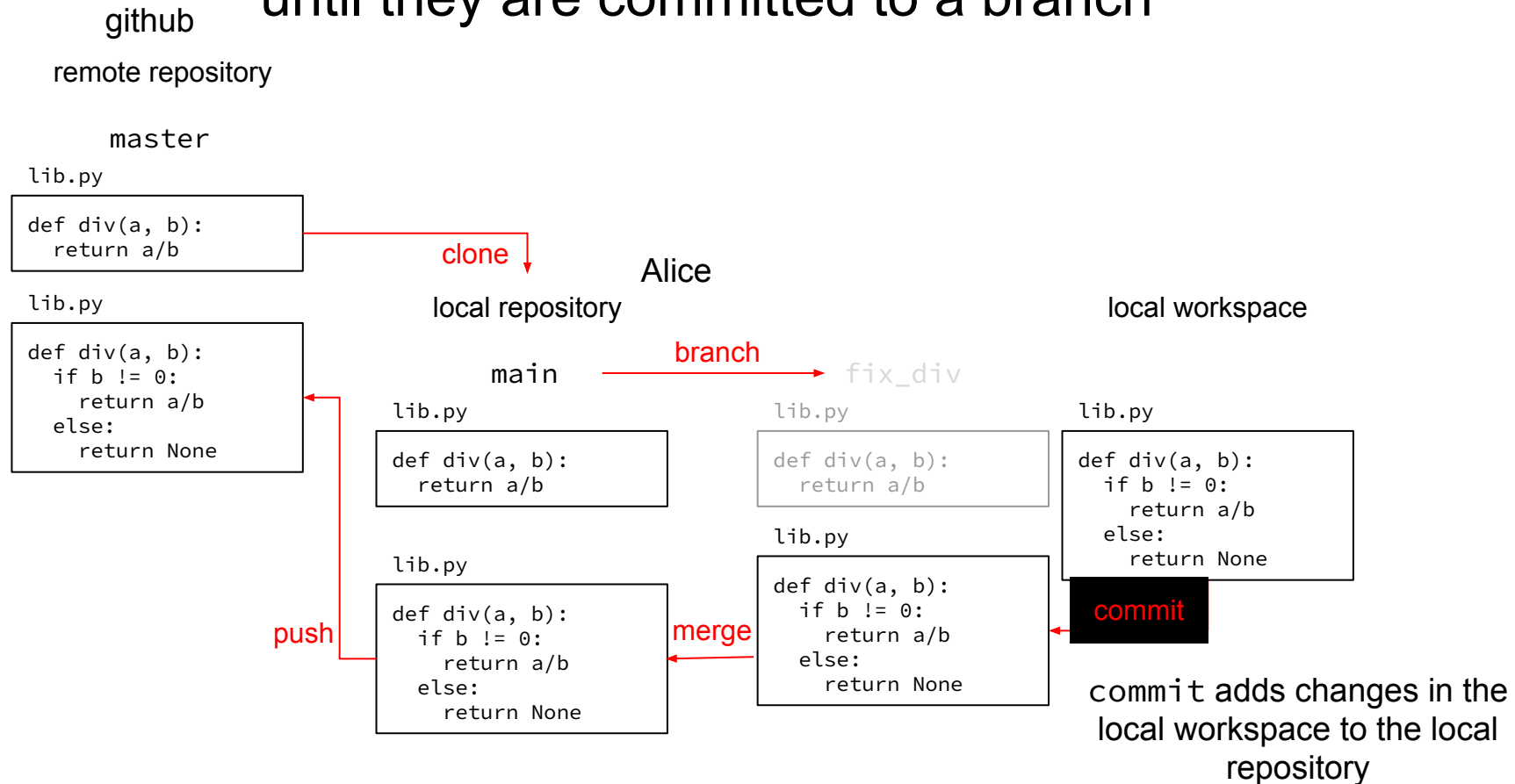
push

merge

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

Edits and new files will stay in your local workspace until they are committed to a branch



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b
```

Alice

local repository

main

branch

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b
```

Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

branch

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b
```



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + a
```

Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + a
```

commit



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + a
```

commit



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + a
```

commit



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b
```

Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

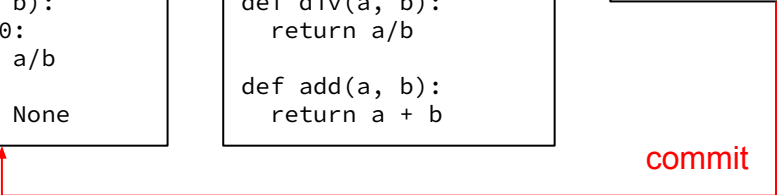
```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

local workspace

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

commit



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

Alice

local repository

local workspace

main

fix_div

add_add

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

merge



Alice

local repository

local workspace

main

fix_div

add_add

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b
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lib.py

```
def div(a, b):  
    return a/b
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lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

merge



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

local workspace

lib.py

```
def div(a, b):  
<<<<<<< HEAD  
    if b!=0:  
        return a/b  
    else:  
        None  
=====  
    return a/b  
def add(a, b):  
    return a+b  
>>>>>> add_add
```

merge



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

local workspace

lib.py

```
def div(a, b):  
    <<<<<<< HEAD  
        if b!=0:  
            return a/b  
        else:  
            None  
    =====  
        return a/b  
def add(a, b):  
    return a+b  
>>>>>> add_add
```

merge



Alice

local repository

main

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

lib.py

```
def div(a, b):  
    if b!=0:  
        return a/b  
    else:  
        None  
  
def add(a, b):  
    return a+b
```

fix_div

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    if b != 0:  
        return a/b  
    else:  
        return None
```

add_add

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

local workspace

lib.py

```
def div(a, b):  
    <<<<<< HEAD  
        if b!=0:  
            return a/b  
        else:  
            None  
    =====  
        return a/b  
def add(a, b):  
    return a+b  
>>>>>> add_add
```

lib.py

```
def div(a, b):  
    if b!=0:  
        return a/b  
    else:  
        None  
  
def add(a, b):  
    return a+b
```

commit

